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# Thermally stimulated currents from corona-charged polypropylene films: A thermal effect of vacuum deposition of metallic electrodes

Atushi Baba and Kazuo Ikezaki

Department of Instrumentation Engineering, Faculty of Science and Technology, Keio University, Yokohama, Japan

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It is demonstrated that the thermally stimulated currents (TSC) from positively or negatively corona-charged polypropylene strongly depend on the order of the following two processes: a process of vacuum deposition of Al electrodes on the sample polymer and a process of heat-treatment of the polymer. Observed results are explained by a thermal effect which is introduced during the vacuum deposition of metallic electrodes. This thermal effect of the vacuum deposition of metallic electrodes is the largest for Al and the smallest for Bi among Al, Au, Ag, and Bi. Observed TSC spectra have three peaks at about 68, 142, and above 147 °C for positively charged samples and four peaks at about 48, 90, 142, and above 147 °C for negatively charged samples, respectively. Origins of these TSC peaks are discussed in some detail.

## I. INTRODUCTION

Thermally stimulated current (TSC) spectra have been widely used for investigation of electrical properties of polymers including electret physics.<sup>1</sup> For measurements of TSC, metallic electrodes such as Al and Au are usually vacuum deposited on sample polymers.

Recently, we demonstrated that the TSC spectra of positively corona-charged polypropylene films markedly depended on the order of the processes of metallic electrode deposition and the heat-treatment of the polymers, and that the observed difference in TSC spectra was attributed to the thermal effect of the metallic electrode deposition.<sup>2</sup>

In this paper, this thermal effect of the electrode deposition on TSC spectra is described in some detail, including the observation of negatively corona-charged polypropylene films.

## II. EXPERIMENTAL PROCEDURES

Undrawn isotactic polypropylene films of 20  $\mu\text{m}$  thickness obtained from Tokyo Serohanshi Co., Ltd. were used for all experiments. In order to study the thermal effect of metallic electrode deposition on TSC spectra, two kinds of samples, type A and type B, were prepared as previously reported.<sup>2</sup> For type A samples polypropylene films were metalized in vacuum on one surface after the films were heat treated at various temperatures, and for type B samples the order of these processes of metallic electrode deposition and the heat treatment was reversed, that is, metallic electrode deposition was performed before the heat treatment of the films. The heat treatment of films was done in such a way that films were maintained at a given temperature for 3 h and then the temperature of the films was reduced to room temperature at a rate of about 1 °C/min or more slowly.

The degree of crystallinity of sample films was estimated from the density of these films which was measured by a floating method with a mixture of isopropyl alcohol and water. The crystallinity of the films was independent of the order of the processes of the heat treatment and the metallic electrode deposition.

Nonmetallized surface of the sample film was positively or negatively corona-charged for 3 min through a dc biased grid electrode which was placed just above the sample film for controlling the initial surface potential of the sample. The initial surface potential was measured by the vibrating electrode method.<sup>3</sup> For most samples, values of the initial surface potential were ranged between 220 and 235 V for positively charged samples and between -215 and -235 V for negatively charged samples, respectively. After measurements of the surface potential, TSCs were observed in open-circuit condition at a heating rate of 3.1–3.4 °C/min. Intensities of TSC presented in figures in the following are not normalized for the initial surface potentials.

## III. EXPERIMENTAL RESULTS

Typical examples of TSC spectra are shown in Fig. 1 for negatively charged type A samples and in Fig. 2 for negatively charged type B samples, respectively. For reference, TSC spectra from positively charged type A and type B samples are shown in Fig. 3. Numerical values of crystallinity and temperature of the heat treatment of sample films are pre-

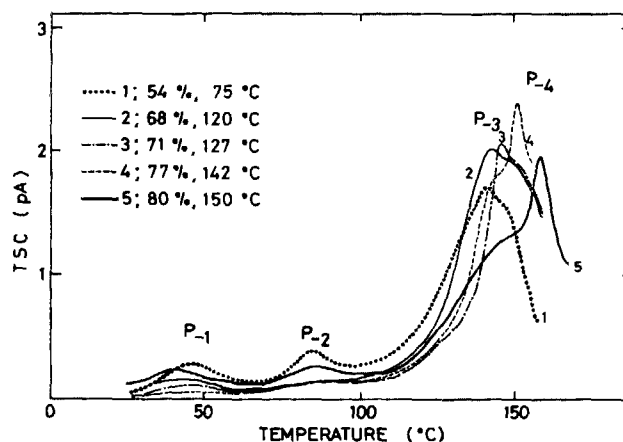


FIG. 1. TSC spectra from negatively corona-charged type A samples with different crystallinities (heat-treatment temperatures).

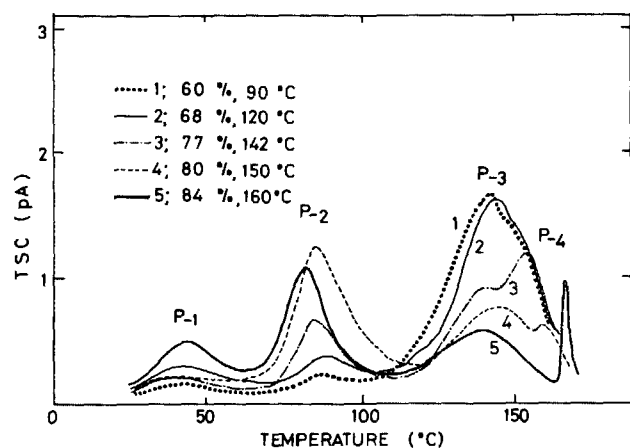


FIG. 2. TSC spectra from negatively corona-charged type B samples with different crystallinities (heat-treatment temperatures).

sented in these figures. As reported previously,<sup>2</sup> three peaks  $P_{+1}$ ,  $P_{+2}$ , and  $P_{+3}$  are observed for positively charged samples.  $P_{+3}$  peak was observed only for samples with high crystallinities, that is, for samples with high treatment temperatures. For positively charged type A samples, peak intensity of  $P_{+1}$  was smaller than that of  $P_{+2}$ . On the contrary, for positively charged type B samples, peak intensities increased for  $P_{+1}$  and decreased for  $P_{+2}$  with increasing the heat-treatment temperature, that is, with increasing crystallinity. For negatively charged samples, four peaks  $P_{-1}$ ,  $P_{-2}$ ,  $P_{-3}$ , and  $P_{-4}$  are observed in their TSC spectra. These four peaks were clearly observed for type B samples. For type A samples, however,  $P_{-1}$  and  $P_{-2}$  were very small and sometimes these two peaks could not be observed at all.  $P_{-4}$  peak was observed only for samples with high crystallinities. With increasing the crystallinity,  $P_{-4}$  shifts its peak position to higher temperatures and increases in its intensity. These observations on TSC spectra from negatively charged samples are very similar to those from positively charged samples. Figures 1 and 2 clearly show that TSC spectra from negatively charged samples are also influenced by the order of processes of the metallic electrode deposition and the heat treat-

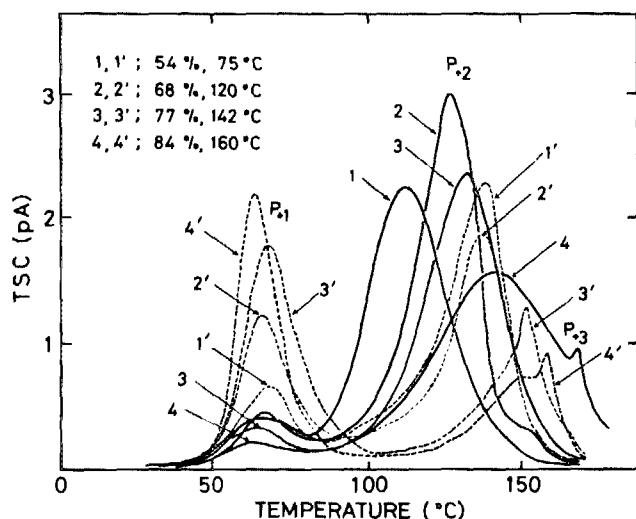


FIG. 3. TSC spectra from positively corona-charged type A (solid line) and type B (broken line) samples with different crystallinities (heat-treatment temperatures).

ment of the sample films. This behavior is the same as the case of positively charged samples.

Relations between the peak temperature and the crystallinity are presented in Fig. 4 for positively charged type A samples, in Fig. 5 for positively charged type B samples and in Fig. 6 for negatively charged type B samples, respectively. For negatively charged type A samples, the relation between the peak temperature and the crystallinity was not examined because of their small peak intensities of  $P_{-1}$  and  $P_{-2}$ . As is seen in Fig. 4, the peak temperature of  $P_{+2}$  for positively charged type A samples scatters in the temperature range between 100 and 150 °C from sample to sample even when the heat-treatment temperature (crystallinity) of the sample films is the same, though the scattering of peak temperature for  $P_{+3}$  of type A samples was smaller than that of  $P_{+2}$  for positively charged type A samples.

Next, we examine the effect of electrode materials on the TSC spectra for Al, Au, Ag, and Bi. Figures 7 and 8 show the observed TSC spectra from positively and negatively corona-charged polypropylene films with different metallic electrodes. Relative intensities of  $P_{-2}$  or  $P_{-1}$  to  $P_{-3}$  and those of  $P_{+1}$  to  $P_{+2}$  are the largest for Bi electrodes and the smallest for Al electrodes. For negatively charged samples with Au electrodes, peak separation between  $P_{-2}$  and  $P_{-3}$  is poor and a broad band extended from  $P_{-2}$  to  $P_{-3}$  is observed in their TSC spectra.

#### IV. DISCUSSION

It is well known that the position and height of TSC peak due to space charge motions depend on the electrode materials.<sup>4</sup> This electrode effect is usually attributed to different efficiencies of charge injection or charge neutralization to metal-polymer interface owing to differences in work function of electrode materials.

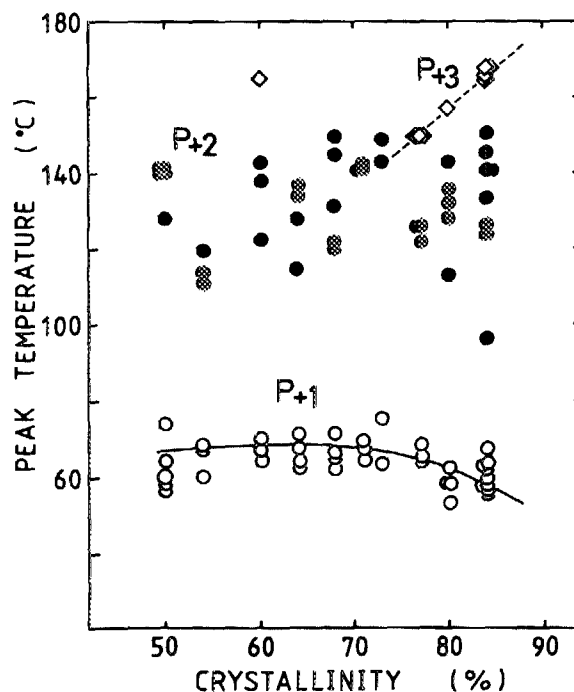


FIG. 4. Crystallinity dependence of peak temperatures of TSC from positively corona-charged type A samples.

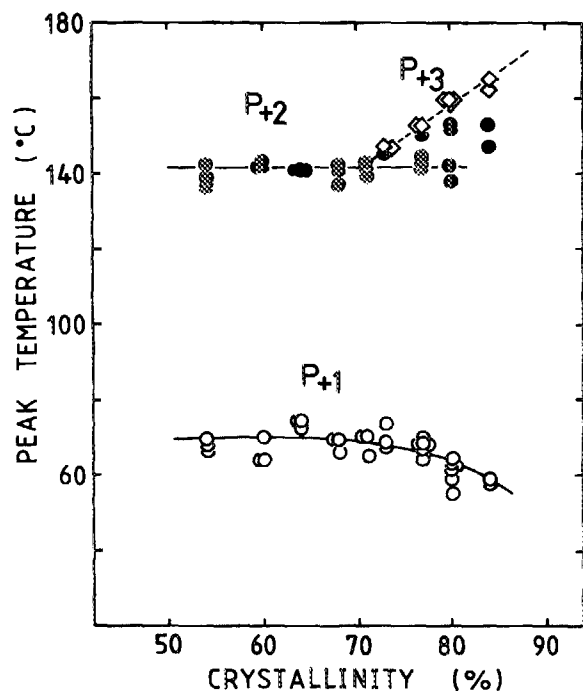


FIG. 5. Crystallinity dependence of peak temperatures of TSC from positively corona-charged type B samples.

In our case, however, the height of TSC peak does not seem to have any relation to the work function of electrode materials (see Table I). In addition to this fact, the polarity effect can not be observed: the differences in TSC spectra

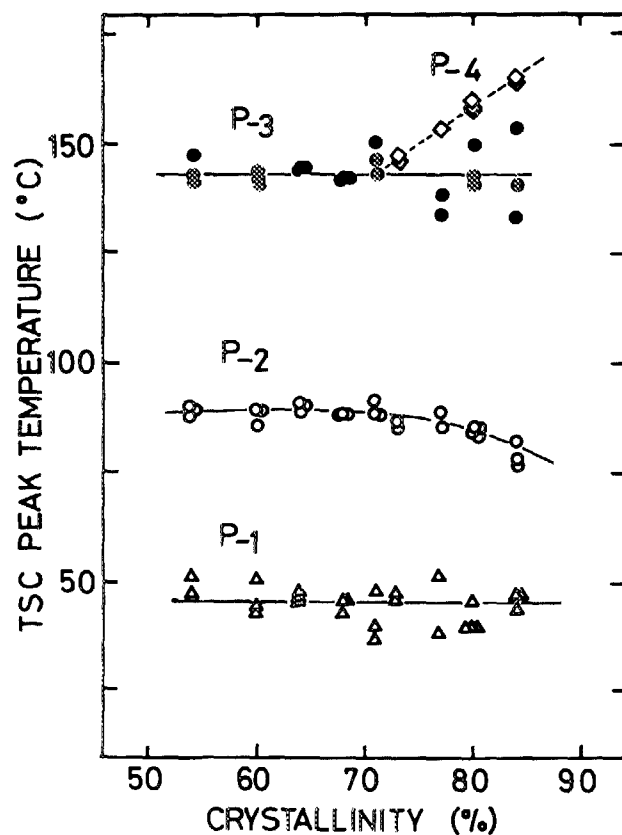


FIG. 6. Crystallinity dependence of peak temperatures of TSC from negatively corona-charged type B samples.

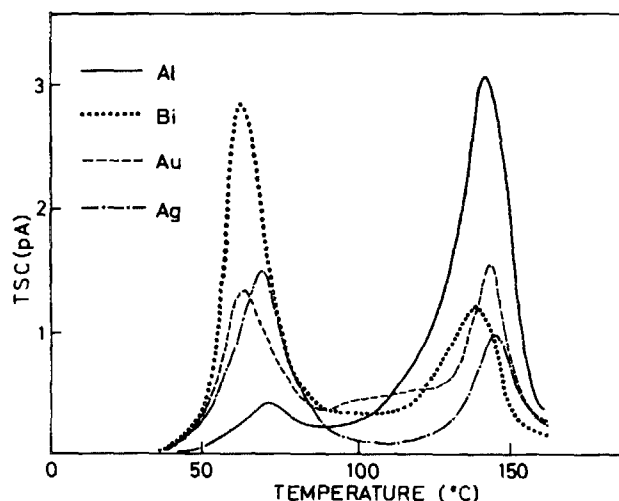


FIG. 7. TSC spectra from positively corona-charged samples with different metal electrodes. Sample films are original and not heat treated, and the crystallinity of these films is 50%.

between type A and type B samples and the electrode material dependence of TSC spectra are observed for both positively and negatively charged samples. These experimental results clearly show that the observed variation of TSC spectra is attributed not to the properties of the metal-polymer interface but to the bulk properties of the sample polymers.

Belous and Wayman<sup>5</sup> have observed the actual surface temperature of an Au film during the deposition of Au on substrates at various temperatures. According to their result, the surface temperature of an Au film deposited at a rate of 10 nm/sec on the substrate at about 300 K attains about 700 K. Main factors of the temperature rise of the deposited metallic films are considered to be (1) the heat radiation  $Q_r$  from an evaporator which is heated at high temperatures, (2) the heat of condensation  $Q_c$  on the basis of phase change, and (3) the kinetic energy  $Q_k$  of evaporated atoms. For most metals  $Q_k/Q_c$  is less than 0.1 and practically  $Q_k$  is negligibly small.<sup>6</sup> Because  $Q_r$  depends on many factors such as the kind of metals which are evaporated, and geometric factors (distance between the evaporator and a substrate, shape and dimensions of the evaporator,....), an accurate estimation of  $Q_r$  is very difficult. A typical estimation of  $Q_r$  shows that  $Q_r$  is less than  $Q_c$  and a main factor of the temperature rise is the

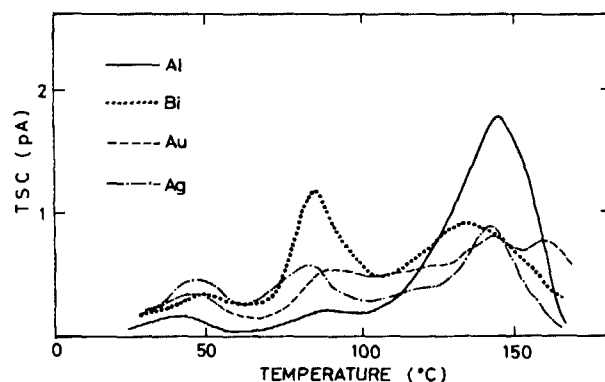


FIG. 8. TSC spectra from negatively corona-charged samples with different metal electrodes. Sample films are original and not heat treated, and the crystallinity of these films is 50%.

TABLE I. Temperature rise due to vacuum deposition of metallic electrodes and the related physical quantities of these materials.

|    | Evaporation temperature <sup>a</sup><br>(°C) | Latent heat of evaporation <sup>a</sup><br>(kcal/g) | Work function <sup>b</sup><br>(eV) | Maximum temperature<br>(°C) |
|----|----------------------------------------------|-----------------------------------------------------|------------------------------------|-----------------------------|
| Al | 1220                                         | 3.1                                                 | 4.2                                | 65–67                       |
| Au | 1400                                         | 0.43                                                | 5.3                                | 46–53                       |
| Ag | 961                                          | 0.68                                                | 4.3                                | 44–52                       |
| Bi | 572                                          | 0.24                                                | 4.3                                | 26–28                       |

<sup>a</sup>T. Sawaki, in *Shinku Jyouchaku (Vacuum Deposition)* (Nikkan Kogyo Shinbunsha, Tokyo, 1968), p. 7.

<sup>b</sup>*American Institute of Physics Handbook* 3rd ed., edited by D. E. Gray (McGraw-Hill, New York, 1972), p. 9–173.

heat of condensation  $Q_c$  for Al so long as the deposition rate is more than  $10^{-2}$  nm/sec.<sup>7</sup> In our case the heat radiation from the evaporator is hardly absorbed by a polypropylene film of 20  $\mu$ m thickness because the temperature measured by a thin film thermocouple which is placed above the empty evaporator, hardly changed irrespective of whether or not a polypropylene film of 20  $\mu$ m thickness was inserted between the thermocouple and the empty evaporator. The main factor of the temperature rise in our case is probably the latent heat of condensation.

In the process of deposition of metallic electrodes onto the polypropylene films, evaporated metallic atoms impinge the polymer surface and condense on it. The molecular motions of the polymer adjacent to these impinging atoms are strongly agitated by the heat of condensation of metallic atoms. The “equivalent temperature” of this molecular agitation seems to be fairly high, but these agitated molecular motions are quenched and frozen in because of the rapid transfer of their energy to the surrounding molecules. Accordingly, the sample film, at least a part of the film which is adjacent to the deposited metallic electrodes, undergoes some partial changes in polymeric structure. Therefore, the films of type A samples are considered to be thermally distorted by the “heat” due to the deposition of metallic electrodes. This thermal distortion in polymeric structure affects the trap sites in the sample films. On the contrary, for type B samples this thermal distortion by the electrode deposition is considered to be annealed or partially annealed depending on the treatment temperature during the heat treatments to change the crystallinity. It may also be attributed to this annealing effect that the scattering of the peak temperature of  $P_{+2}$  for type B samples became smaller than that for type A samples as shown in Figs. 4 and 5.

Previously we reported<sup>2</sup> that the observed differences in TSC spectra from positively corona-charged type A and type B samples can be attributed not to the different efficiencies of charge injection or charge neutralization at metal-polymer interface but to the thermal distortion of polymers by the deposition of metallic electrodes, TSC spectra from negatively corona-charged type C samples were examined in the same way as the case of positively charged samples which was reported previously. Type C samples were prepared as follows: a thin layer of Al was vacuum deposited once again on the metalized surface to type B samples. Therefore, the films of type C samples are considered to have again a ther-

mal distortion. The observed TSC spectrum is shown in Fig. 9 for a negatively charged type C sample of which the crystallinity is 80%. In Fig. 9, TSC spectra from negatively charged type A and type B samples with the same crystallinity to the type C samples are presented for reference. In the TSC from the negatively charged type C samples, no peaks of  $P_{-1}$  and  $P_{-2}$  can be observed. This TSC spectrum is very similar to the spectra from type A samples. It is obvious from this experimental result that TSC spectra from negatively corona-charged polypropylene films are markedly influenced by the thermal distortion of the polymer due to the vacuum deposition of metallic electrodes.

It is considered that metals with large latent heat of evaporation and with high evaporation temperature release a large amount of energy due to the heat of condensation and the heat of radiation from the evaporator. Values of evaporation temperature (temperature at which the vapor pressure of evaporated metallic atoms becomes to be  $10^{-2}$  Torr), latent heat of evaporation (including the energy which is required to heat the metal from room temperature to its evaporation temperature) and work function of the metals used in this study are tabulated in Table I. In Table I, values of the maximum temperature of sample polypropylene film which are observed during the deposition of metallic electrodes, are also presented. These temperature rises were measured by Bi-Sb thin film thermocouples which were prepared by the vacuum deposition of Bi and Sb directly on the back surfaces of sample polypropylene films. The heat capacity of the thin film thermocouple is about  $4.0 \times 10^{-5}$  J/K per unit area. This value of the heat capacity is negligibly small in comparison with the heat capacity of sample polypropylene film,  $6.3 \times 10^{-3}$  J/K per unit area. From Table I, the energy release at condensation is considered to be the largest for Al and the smallest for Bi among the four metals. The latent heat of evaporation for Bi is less than a tenth of that for Al. This fact well corresponds to the observed values of the temperature rise in Table I and the TSC behavior which can be seen in Figs. 7 and 8: the relative intensities of  $P_{+1}$  to  $P_{+2}$  for positively charged samples and  $P_{-1}$  or  $P_{-2}$  to  $P_{-3}$  for negatively charged samples are the largest for Bi electrodes and the smallest for Al electrodes. It is concluded from these experi-

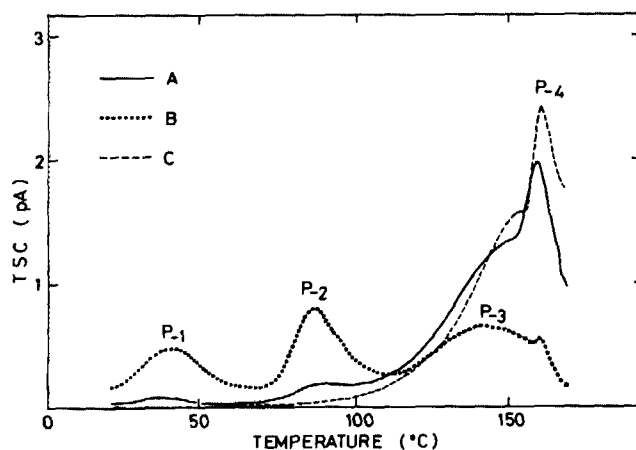


FIG. 9. TSC spectra from negatively corona-charged type A, B, and C samples. The crystallinity of these three kinds of sample films is 80%.

mental results that the thermal effect of the vacuum deposition of metallic electrodes on TSC spectra is the smallest for Bi among the four metals. This conclusion is also supported by the fact that TSC spectra from samples with Bi electrodes resemble those from samples with silver paste electrodes which can not have any thermal effect of TSC spectra. One of the examples of TSC spectra from negatively charged samples with the silver paste electrodes is presented in Fig. 10. As is seen in Fig. 10, this TSC spectrum is very similar to that from samples with Bi electrodes in the temperature range below about 130 °C. Abrupt decrease and current reversal of this TSC observed above 130 °C is considered to be due to the destruction of the silver paste electrode and the sample film itself by an action of solvent in the paste at high temperatures.

Because of the large latent heat of Al, the thermal effect of the vacuum deposition of Al electrodes is large even when the deposition rate is very slow. Figure 11 shows TSC spectra from positively charged polypropylene films with Al electrodes which were prepared at different deposition rates. Curve A, B, and C in Fig. 11 are the TSC spectra corresponding to the following Al deposition conditions (1), (2), and (3), respectively. The deposition rate was changed by changing the distance  $d$  between the evaporator and the sample film. Aluminum electrodes of about 100 nm thickness were prepared under the following three different evaporation conditions. (1) The distance  $d$  is 6 cm and the deposition time is about 15 sec. Without otherwise mentioned Al electrodes were prepared under this condition for all samples used in TSC measurements. (2) The distance  $d$  is 13.5 cm and the deposition time is 90 sec. (3) The distance  $d$  is 13.5 cm and the electrode deposition is intermittently performed: after the deposition for 3 sec a shutter is closed to prevent the deposition for an interval of 10–15 sec. This intermittent deposition was repeated 30 times to prepare Al electrodes of about 100 nm thickness. As shown in Fig. 11, TSC spectra from these three samples are almost the same. They are characteristic of type A samples: the peak intensity of  $P_{+1}$  is much smaller than that of  $P_{+2}$ .

Although different deposition rates result in different temperature rises of the substrate polypropylene film, TSC spectra from samples with Al electrodes hardly depend on the deposition rate of Al electrodes as shown in Fig. 11. The temperature rise produced by the Al electrode deposition is

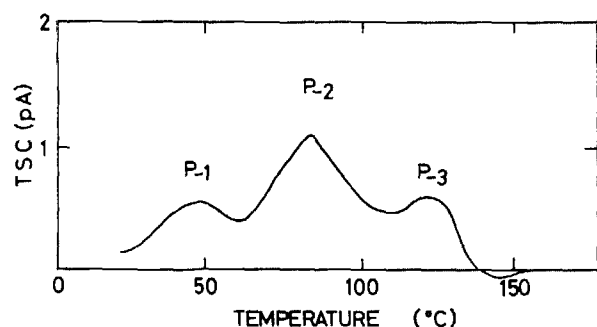


FIG. 10. TSC spectrum from a negatively corona-charged polypropylene film with a silver paste electrode. The film is original and not heat treated, and the crystallinity of the film is 50%.

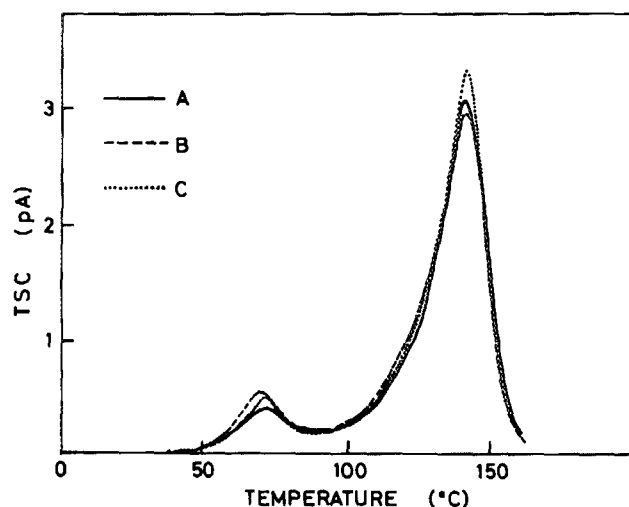


FIG. 11. TSC spectra from positively corona-charged polypropylene films with Al electrodes which were prepared by different evaporation conditions: A is prepared by the evaporation condition (1), B the evaporation condition (2), and C the evaporation condition (3). Details of these evaporation conditions are described in the text.

about 45 °C at most even in the deposition condition (1) because the temperature of a room where the temperature rise measurements were made was 23 °C. This temperature rise apparently seems to be too low to explain the observed drastic variation of TSC spectra. However, this low temperature rise is not inconsistent with the early explanation for the thermal effect of the vacuum deposition of metallic electrodes because the temperature which is measured by the thin film thermocouple is a kind of the averaged temperature over the whole sample film. In addition to this fact, this averaged temperature rise of the substrate polymer is not a cause but an effect of a distortion of polymeric structure due to the deposition of metallic electrodes. Therefore, molecular distortion does not always depend on the averaged temperature rise of the substrate polymer. It rather depends on the local temperature rise of the polymer when the evaporated metallic atoms condense on the surface of the polymer.

Finally, we briefly discuss the origin of TSC peaks observed in corona-charged polypropylene. Many authors have reported TSC spectra from polypropylene under different experimental conditions such as sample preparation (including electrode preparation), charging conditions or TSC measurement conditions.<sup>8–16</sup> In these TSC studies, two kinds of broad TSC bands, a high temperature band which has a peak at around 140 °C and a low temperature band at about 80 °C, are reported above room temperature, though Jarrigeon *et al.*<sup>8</sup> and Pillai *et al.*<sup>9</sup> separate several TSC peaks from these broad TSC bands. As for the high temperature TSC band which has a peak at about 140 °C, there are only studies by Takamatu and Fukada,<sup>10</sup> and Motyl.<sup>11</sup> Takamatu and Fukada studied TSC with 0.2-mm-thick thermoelectrets which were prepared from powder polypropylene by thermal compression and charged with thin foil mechanical contact electrodes. They measured TSC under the open-circuit condition. They observed the high temperature TSC band and a sharp TSC peak at 165 °C, but they did not observe the low temperature TSC band. On the contrary, Mo-

tyl observed the both of the high temperature band and the low temperature band, but not the sharp TSC peak at high temperature. He studied the polypropylene TSC with positively corona-charged films of 26  $\mu\text{m}$  thickness under various electrode conditions.

Although rigorous comparison between these reported TSC spectra can not be made because of their different experimental conditions, TSC peaks of  $P_{+3}$  and  $P_{-4}$  in our study correspond to the sharp TSC peak observed by Takamatu and Fukada.<sup>10</sup> Peak temperature of  $P_{+3}$  and  $P_{-4}$  are presented in Fig. 12.

It is well known that the heat treatment of crystalline polymers such as polyethylene and polypropylene results in increase of the x-ray long period or lamellar thickness depending on the heat-treatment conditions such as the treatment temperature and time,<sup>17,18</sup> and that the melting point of the polymer becomes high with increasing of the lamellar thickness and of the crystal perfection.<sup>19</sup> Therefore, the heat treatment of our polypropylene films is considered to shift the melting point towards higher temperature side getting close to its equilibrium melting point. As is seen in Fig. 12, peak temperatures of  $P_{+3}$  and  $P_{-4}$  peak increase with the crystallinity, that is, with the temperature of the heat treatments, and their extrapolated values at 100% crystallinity become very close to the equilibrium melting point 186 °C (Ref. 20) of polypropylene. Figure 12 also shows that the crystallinity dependence of the peak temperatures of  $P_{+3}$  and  $P_{-4}$  is expressed by the same relation irrespective of the type of the samples or the kind of the electrode materials. This means that the trap sites related to  $P_{+3}$  and  $P_{-4}$  peaks seem to be stable against the thermal distortion introduced by the vacuum deposition of metallic electrodes. From these facts it is concluded that  $P_{+3}$  and  $P_{-4}$  peaks are attributed to charge carriers in deep traps such as defects in a crystalline region. These charge carriers are mobilized by melting of the crystalline region.

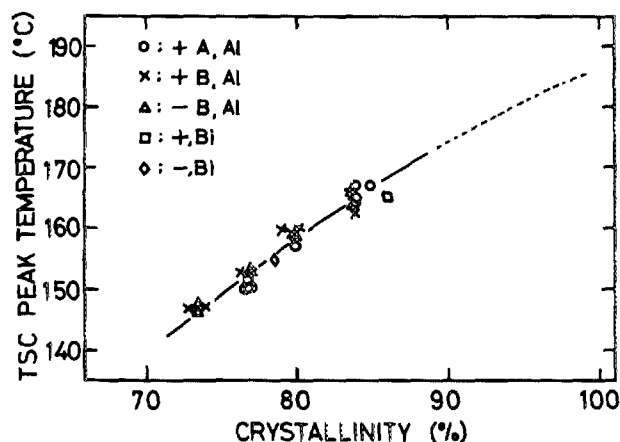


FIG. 12. Crystallinity dependence of peak temperatures of  $P_{+3}$  and  $P_{-4}$  for various samples: positively corona-charged type A (+ A, Al) and type B (+ B, Al) samples with Al electrodes, negatively corona-charged type B samples with Al electrodes (- B, Al) and positively corona-charged (+, Bi) and negatively corona-charged (-, Bi) samples with Bi electrodes. For samples with Bi electrodes, vacuum deposition of Bi electrodes was made after the heat treatment of sample films. Therefore, these samples correspond to type A samples with Al electrodes.

The high temperature bands,  $P_{+2}$  and  $P_{-3}$  in our study are observed in the temperature range from 100 to 160 °C. The peak temperatures of these bands are 142–3 °C independent of the crystallinity though the values of this peak temperature scatter at high crystallinities. This scattering may be due to poor peak separation of these peaks from the sharp peaks  $P_{+3}$  and  $P_{-4}$ .

Recently it is reported that the lamellar thickness of cold-drawn isotactic polypropylene rapidly increases by annealing the polymer at temperatures above 100 °C.<sup>21</sup> It is also reported that polymer chain ends on the surface of lamellae move into the crystalline region at high temperatures.<sup>22</sup> This including the chain end in the lamellae of isotactic polypropylene is more enhanced at temperatures above 147 °C.<sup>23</sup> These results suggest that heating of polypropylene in the temperature range in which  $P_{+2}$  and  $P_{-3}$  TSC peaks are observed, induce molecular motions on a fairly large scale in the crystalline phase. Therefore, the origin of  $P_{+2}$  and  $P_{-3}$  is thought to be the charge carriers which are released by these molecular motions from traps such as crystalline defects at or near the surface of crystallites.

Although peak temperatures of the high temperature band are almost independent of the charging polarity, the peak temperature of the low temperature band depends on the charging polarity. Positively charged samples have one TSC peak at 68 °C in the low temperature region, while negatively charged samples have two separate TSC peaks at 90 and 48 °C. These low temperature TSC peaks are very sensitive to the thermal history of the samples: the peak intensity of these TSC peaks is strongly affected by the vacuum deposition of metallic electrodes.

In polypropylene double glass transitions have been reported: the upper one  $T_g$  (U) and the lower one  $T_g$  (L).<sup>24,25</sup> Temperatures for these transitions vary depending on the crystallinity. Typical values of  $T_g$  (L) and  $T_g$  (U) are -19 and 18 °C for samples with the crystallinity of about 55%.<sup>24</sup> These values increase with crystallinity.

Jarrigeon *et al.*<sup>8</sup> separated several “elemental” TSC peaks from the observed broad TSC spectra for undrawn and drawn polypropylene thermoelectrets. They attributed “elemental” peaks which were observed for undrawn polypropylene at -8 and 29 °C, to the double glass transitions  $T_g$  (L) and  $T_g$  (U), respectively. They also reported that 39 and 68 °C peaks were attributed to liquid-liquid transitions (transitions from one liquid phase to another). A phase transition from smectic to monoclinic structures at about 50 °C has also been reported.<sup>26,27</sup> As shown in Fig. 6, peak temperature of  $P_{-1}$  is close to the reported temperature values of transitions in polypropylene such as the upper glass transition, the liquid-liquid transition and the smectic to monoclinic transition. Observed values of  $P_{-1}$  peak temperature seem to be independent of the crystallinity. Peak temperatures of  $P_{+1}$  and  $P_{-2}$  are much higher than that of  $T_g$  (U) and they slightly decrease at high crystallinities as shown in Figs. 5 and 6. Therefore, the origin of these TSC peaks  $P_{-1}$  and  $P_{-2}$  and  $P_{+1}$  which appear in the low temperature region in our study may be thought to have no direct relation to the upper glass transition. At this present stage, it is not clear the  $P_{-1}$  and  $P_{+1}$  peaks are related to the liquid-liquid transitions at 39

and 68 °C, respectively. The peak  $P_{-1}$ , probably has no relation to the transition from smectic to monoclinic because this phase transition occurs in quenched polypropylene while the  $P_{-1}$  peak appears in "annealed" samples. One of the possible origins for these TSC peaks which appear in the low temperature region in our study, is an intermediate phase such as a restraint region between crystalline and amorphous phase.

In conclusion, the vacuum deposition of metallic electrodes seriously affects TSC spectra when (1) the sample is a crystalline polymer with a relatively low melting points such as polypropylene used in this study, (2) the sample polymer is of thin film, and (3) the electrode metals release a large amount of energy when they are deposited. From various experimental results, it can be said that Bi is the most suitable electrode material for TSC measurements among the four metals, Al, Au, Ag, and Bi so long as TSC measurements are performed in the temperature range below about 180 °C.

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